MIL-D-21201F
9 August 1984
SUPERSEDING
MIL-D-21201E
30 June 1976

MILITARY SPECIFICATION

DRILL, PNEUMATIC DRIFTER, SELF-PROPELLED, CRAWLER MOUNTED (TRACK DRILL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers a self-propelled, crawler mounted, drifter drill (referred to hereinafter as "track drill"), having a minimum 4-3/4-inch cylinder bore.

1.2 Classification. The track drill shall utilize one of the following types for drill rotation as specified (see 6.2):

Type I - Rifle bar.
Type II - Independent air motor drive.

2. APPLICABLE DOCUMENTS

2.1 *Government documents.*

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-T-3351 - Tractor, Full-Tracked, Low Speed; Tractor Wheeled, Agricultural; and Tractor Wheeled, Industrial; and their Attachments, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3820

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MS51335 - Pintle Assembly, Towing, 18,000 lbs Capacity, Manual Release.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

S1.4 - Specification for Sound Level Meters.
S1.13 - Methods for the Measurement of Sound Pressure Levels.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The track drill shall essentially consist of a crawler chassis with a pintle attachment for towing an air compressor, air motors, dust-suppression system, hydraulically operated swing type boom, chain-fed type drill retainer channel, and an independent air motor or rifle bar rotation type pneumatic drifter drill. The track drill shall be provided with lubrication fittings, hoses, controls, crawler brakes or locks, and other necessary items to perform complete and efficient drifter drill operations. The track drill shall use compressed air from the towed air compressor as a power source for propulsion and drilling.

3.1.1 Standard commercial product. The track drill shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the track drill being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified (see 6.2), the contractor shall furnish a track drill for first article inspection and approval (see 4.2.1 and 6.4).

* 3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Safety. All rotating or reciprocating parts, excluding the drill steel, feed chain, and crawler track assembly, that are hazardous to operating personnel shall be enclosed or guarded.

3.4.1 Noise level. Unless otherwise specified (see 4.4.9 and 6.2), when equipment noise level exceeds the 85 db level for the required 8 hour exposure level of OSHA Health and Safety Standard (Title 29 CFR 1910.95) at the operator's position, measured in accordance with ANSI S1.13 on the A scale of a sound level meter conforming to ANSI S1.4, a warning plate shall be attached to the track drill in a location conspicuous from the control panel stating that hearing protection is required when operating the track drill.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

3.5 Ease of maintenance. All major assemblies and installed attachments shall be readily accessible for servicing, repair, or replacement. Drainage of lubricants and hydraulic fluid shall be to the ground without draining on or over any part of the track drill. When drain outlets are located over parts of the track drill, integral tubes or pipes shall convey the fluid from the drain to the ground. Drain outlets shall be accessible without the removal or adjustment of accessories or parts other than access opening covers.

3.6 Performance. The drifter drill, employing a tungsten carbide button type bit, operating at 90 pound-force per square inch (psi) rated air pressure and using air at a rate not to exceed 600 cubic feet per minute (cfm) for type I drifter drive and 750 cfm for type II, shall drill not less than a 3-1/2-inch diameter hole, 12 inches deep in 1 minute when operating in Barre granite or equal with a crushing strength of 17,000 psi. When crushing strength of the granite is other than 17,000 psi, the penetration required shall be inversely proportional to the penetration required at a crushing strength of 17,000 psi. The track drill shall drill 30-foot deep holes in homogeneous rock with an average specific gravity of not less than 2.6, with the boom and mast at various angles and positions for 50 hours of throttle time without replacement of parts other than drill steel and bits. The track drill shall operate on 35 percent side slopes, or 50 percent longitudinal slopes, or any combination thereof without either track losing all contact with the ground.

* 3.7 Crawler chassis. The track drill shall be mounted on a self-propelled crawler chassis consisting of two oscillating crawler tracks. The crawler chassis shall have a ground clearance of not less than 10 inches. The crawler mounting shall permit tracks to rise over obstructions while maintaining a proportionately level platform when traveling or drilling. The vertical oscillation of the track frame shall be not less than 14 inches, measured at the center of the front idler. Suitable track rollers shall be provided to assure smooth track operation and maintain even ground pressure. Heavy-duty recoil springs shall be furnished for cushioning front idlers. Track rollers and the track rail shall be arranged to prevent damage to bearings or grease fittings when the track drill is operated through abrasive material or loose stone. The track drill shall be equipped with track adjusters for maintaining tension on the track. Track frames shall not be spliced. The minimum track gage shall be 70 inches from center to center of tracks and a ground bearing length of not less than 60 inches.

3.7.1 Track chain. Track links and grouser shoes shall be heat-treated steel of drop-forged or roller construction, and all similar parts shall be interchangeable. The shoes shall be not less than 10 inches wide with provisions for affixing rubber crawler-tread pads. Track links shall be equipped with replaceable pins and bushings. The master pin shall be secured by a lock or a press fit and shall be identified.

3.7.2 Tramming air motors. A tramming air motor using a gear drive train or chain drive, shall drive each track. Each motor shall have a spring-loaded deadman control for forward and reverse movement of the track. The motor and drive train shall be enclosed. Adequate vent p[illegible]gs shall be provided to prevent air pressure buildup in track drive housing. The rated horsepower of the air motors at 90 psi pressure shall be not less than that required to

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

advance the track drill up a grade of not less than 20 percent for 100 yards, and over level ground at a speed of not less than 1.3 miles per hour (mph), while towing the air compressor unit weighing not less than 8,500 pounds (lb). The air motors shall have means for being disengaged from the crawler drive train without the use of tools to permit the track drill to be easily towed.

3.7.3 Brakes. Each track shall be provided with brakes or track locks to prevent movement of the track drill when on a 50 percent longitudinal slope. Track locks or brakes shall be operable when power is lost.

3.7.4 Platform. When specified (see 6.2), a platform for use by the track drill operator, while manipulating the tramming controls during road travel, shall be furnished.

* 3.8 Boom. The boom shall be raised, lowered, and swung hydraulically. Vertical movements of the boom shall be such that holes may be drilled anywhere within the range of from 2-1/2 to 8-feet above the ground surface of the tracks. Movement of the boom shall be such that a vertical hole may be drilled on a line not less than 12 inches outside of either track without repositioning the crawler. The drifter drill shall be capable of drilling regardless of the position of the boom and mast. Boom movement shall be so controlled that the boom can be stopped and fixed without creepage at all points in its range of vertical travel.

* 3.9 Mast. A mounting plate shall be used to secure the fabricated steel mast to the boom. The mounting plate shall be hydraulically operated and shall be capable of sliding on the mast for a minimum distance of 3-feet. A reversible air motor or hydraulic cylinder(s), secured to the mast, shall be used to drive the feed mechanism. The mast assembly shall include an adjustable feed chain, upper and lower feed chain idler sprockets, a drifter drill mounting plate, a foot pad, and a drill steel centralizer. Feed chain and idler sprockets shall be arranged so that total travel of the drifter drill shall allow the operator to change 12-foot sectional drill steel and couplings, by disengagement of the centralizer only. When operating at an air pressure of 90 psi, the air motor or hydraulic cylinder(s) shall drive the feed chain, and shall provide a maximum effort of not less than 3,000 lb for crowding and retracting the drifter drill. The mast assembly shall be capable of a hydraulic roll operation through an arc of not less than 70 degrees (°) from right to left. The vertical mast dump from a horizontal position shall have a minimum travel of 145° with not more than one mechanical pin change. The mast movement shall be controlled so that the mast can be stopped and fixed without creepage at all points in its range of travel. The mast shall be of such strength to support the drifter drill and withstand the imposed loads.

* 3.10 Pneumatic drifter drill. The pneumatic drifter drill shall have a bore size of not less than 4-3/4 inches, shall operate on a rated air pressure of 90 psi, and shall be automatically valved to provide not less than 1,500 blows per minute to the chucked drill shank. The type I drifter drill shall weigh not less than 160 lb and shall be of the long stroke hammer type. The type II drifter drill shall weigh not less than 280 lb and shall be of the high speed, short stroke type with independent rotation. The chuck shall be designed to prevent the shank from leaving the drifter drill, and shall contain parts of high wearing quality. The chuck shall accommodate a four spline drill

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

shank with rope thread. Unless otherwise specified (see 6.2), the track drill shall be furnished with 5 drill shanks. The drifter drill shall also provide for control of clockwise, neutral, and counterclockwise rotation of the drill rod. The type II drifter shall also provide drill rod rotation independent of hammer control. The drifter drill shall be designed for wet or dry drilling, and to allow water or air to pass through the unit and the attached drill rod for flushing or blowing purposes. A lubricator shall be provided in the air line and have an oil reservoir of sufficient capacity for not less than 8 hours of operation. The reservoir shall be equipped with a filler opening with a removable strainer. The cap or reservoir immediately adjacent to the filler cap shall be plainly marked to indicate "Rock Drill Oil." The lubricator shall give thorough and positive lubrication to the drifter drill and feed motor at all times regardless of the position of the drifter drill.

3.11 Controls. All controls shall be clearly marked as to use and function. The crawler, boom, and mast controls shall be grouped for safe operation and ease of operation when positioning the crawler, boom, and mast preparatory to drilling. The drilling controls shall be grouped for safe operation and shall be accessible to the operator when drilling at any position. Drilling controls shall operate the drifter drill functions including the rotation, feed, and flushing operations.

3.12 Pintle. A pintle assembly, conforming to MS51335, shall be provided to receive the tongue of a 600 or 750 cfm air compressor.

3.13 Hoses. Interconnecting hose lines shall be held to a minimum, consistent with complete operation and shall be provided with nonleak connections. Hoses shall be free from pits, blisters, checks, or other defects and shall be of uniform thickness. Hoses shall be designed for a minimum 200 psi working pressure and shall pass a 600 psi hydrostatic test pressure. Pneumatic hose shall have an oil resistant inner tube and scuff resistant outer cover. Pneumatic hose shall be supported, where practicable, to prevent kinking. One connection shall be provided to receive the supply hose line from the compressor and shall be 2 inches with 11-1/2 threads per inch National taper pipe (thread), external thread as specified in FED-STD-H28.

3.14 Hydraulic system. The hydraulic system shall be protected against the entrance of dirt or other foreign materials and shall be complete with all operating accessories, including a pump, hydraulic cylinders, reservoir, pressure relief valve, and an oil filter or oil strainer accessible for cleaning. The pump shall be of the gear or vane type driven by an air motor. The pump shall supply a continuous flow of hydraulic fluid, and shall have sufficient capacity to energize all hydraulically powered components. Hydraulic cylinders shall be of the double-acting type having lock-type check valves. Hydraulic valves shall control retracting speed of all hydraulic cylinders, or an internal flow restriction or bypass shall be provided at each end of the cylinders. A wiper shall be provided on the cylinders, adjacent to the piston rod seal, to prevent dirt and abrasives from damaging the seal or entering the cylinder.

3.15 Dust-suppression system. Unless otherwise specified (see 6.2), a dust suppression system shall be provided. The dust-suppression system shall be protected against the entrance of dirt or other foreign materials. It shall be complete with all operating accessories, including a suitable steel

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

reservoir, automatic air operated valve that turns on liquid flow when blowing air is turned on, adjustable fine control, a filler-opening cap, and hoses. The steel reservoir shall be of sufficient size to hold the quantity of detergent water solution necessary to insure 8 hours of operation of the equipment without being refilled. There shall be no leakage of fluid when the system is operated.

3.16 Toolbox. A toolbox shall be provided to carry those tools and accessories required to operate the unit. The toolbox shall have a hinged lid and trunk drawbolt of a type that will keep the lid closed when the toolbox is subjected to vibration. The toolbox shall be securely fastened to the unit in a protected and accessible location.

3.17 Lifting and tiedown attachments. The track drill shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the track drill. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments shall be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the track drill on the carrier when shipped.

* 3.18 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be *painted shall be cleaned and dried to insure that they are free from* contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.19 Identification plate. An identification plate will be furnished by the contracting officer for each track drill . The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each track drill in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8 inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.20 Instruction plates. The track drill shall be equipped with instruction plates suitably located, describing any special or important

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter. A warning plate shall be conspicuously affixed to the unit immediately adjacent to the reservoir filler cap of the air line lubricator. The plate shall be inscribed as follows:

WARNING

Under normal operation, the line oiler system is pressurized to 90 psi. Do not open filler cap until air pressure has been completely eliminated from the air system.

* 3.21 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.22 Transportability. When specified (see 6.2), the track drill shall withstand impact forces encountered in rail transport and semitrailer transport without damage or permanent deformation.

3.23 Workmanship.

3.23.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.23.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.23.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.23.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.23.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on track drill when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4, and, when specified, the preproduction pack inspection of 4.5 (see 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

* 4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.10, and the packaging inspection of 4.5.

4.3 Examination. Each track drill shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

4.4 Tests.

4.4.1 Test procedures. The first article shall be subjected to the tests specified herein. Where drilling ability is being tested, all drilled holes shall be made with 3-1/2-inch bits and all drilling shall be performed using 12-foot drill rod. Failure to pass any phase of the following tests shall be cause for rejection.

4.4.2 Boom and mast positioning test. With the mast vertical, drill holes 1-foot deep on a line not less than 12 inches outside of each track. With the mast horizontal, drill two face holes, one at a height of 2-1/2 feet and the other at a height of 8-feet measured from the ground surface of the track to the center of the holes.

4.4.3 Towing test. To verify conformance to 3.7.2, the track drill shall be capable of towing an air compressor unit weighing not less than 8,500 lb under the following conditions:

a. When maintaining a travel speed of not less than 1.3 mph over a level 1 mile measured course.
b. When negotiating not less than a 20 percent incline for not less than 100 yards.

4.4.4 Drilling Test. The track drill shall perform the following tests to verify conformance to 3.6. During the drilling tests, the required drilling speed must be maintained and the air consumption must be no more than 600 cfm for the type I track drill and no more than 750 cfm for the type II at 90 psi pressure. The performance of the track drill shall show no significant difference during the tests. The track drill shall be examined for any evidence of damage upon completion of the following tests:

a. The track drill shall drill 30-foot deep holes at various angles and positions through 50 hours of operation. Normal maintenance will be permitted, but no adjustments to the track drill or its component parts shall be allowed. The track drill towing the compressor shall move approximately 10 feet between each hole.
b. Drill five holes 30 inches deep in Barre granite, or other rock of similar hardness and uniform density, at an average drilling speed as specified in 3.6. When a dust-suppression system is required, all drilling tests shall be performed wet to verify conformance to 3.15.

4.4.5 Stability test. With the right side of the track drill downhill, drive the track drill forward 50 feet on a 35 percent side slope. Repeat this procedure with the left side of the track drill downhill with no change in the position of the mast or boom. Then drive the track drill forward down a 50 percent longitudinal slope. Inability of either track to maintain ground contact shall constitute failure of this test.

4.4.6 Brake test. Place the track drill facing downhill on a 50 percent slope and apply the brakes or track locks. This test shall verify conformance to 3.7.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

4.4.7 Rotation test. The drifter drill shall be tested to determine rotation and blows per minute specified in 3.10 for wet or dry drilling.

4.4.8 Lifting and tiedown attachments test. The track drill, when equipped with lifting and tiedown attachments, shall be tested to verify that the attachments conform to the requirements specified in 3.17.

4.4.9 Noise level test. Compliance with the noise level limits of 3.4.1 shall be verified by measurement. Noise levels shall be measured using a sound level meter in accordance with ANSI S1.4, type 1. The meter shall be set for the A-weighting "slow" mode of operation. The noise level test shall be set up and conducted in accordance with the guidelines specified in ANSI S1.13 for the "Field" measurement method.

4.4.10 Operational test. Each track drill produced in fulfillment of a government contract shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The air motors shall be started and subjected to a warmup period as recommended by the manufacturer. The track drill shall be given a run-in test and all controls operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operate promptly, fully, and without restriction or malfunction. The hydraulic system, including hoses, fittings, and seals shall be examined for evidence of leakage or failure. Gear case bearings, shafts, and other moving parts shall be examined for excessive heating or abnormal operation insofar as is practicable. Failure to pass any phase of this test shall be cause for rejection.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-T-3351. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-T-3351 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. This self-propelled crawler mounted drifter drill, when completely equipped for drilling, shall be utilized for horizontal and vertical drilling to define crevice lines, to drill blast holes, and to accomplish other miscellaneous drilling jobs under universal military operating conditions.

*6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Specifications, Standards, and Handbooks (see 2.1.1).
d. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

e. When noise level is other than specified (see 3.4.1).
f. When an operator's platform is required (see 3.7.4).
g. When shanks are not required (see 3.10).
h. When dust-suppresion system is required (see 3.15).
i. Color of finish coat required (see 3.18).
j. When other means for lubrication are required (see 3.21).
k. When transportability is required (see 3.22).
l. When a quality conformance inspection is required (see 4.5).
m. When a preproduction pack inspection is required and time frame required for submission (see 4.5).
n. Level of preservation and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.4 First article. When a first article inspection is required (see 3.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one track drill. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing Activity:

Navy - YD

(Project 3820-0179)

Review Activities:

Army - AT
Navy - MC
DLA - CS
AF - 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

Commanding Officer (04M2)
Naval Facilities Engineering Command
200 Stovall Street
Alexandria, VA 22332-2300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer (156)
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-21201F	DRILL, PNEUMATIC DRIFTER, SELF-PROPELLED, CRAWLER MOUNTED (TRACK DRILL)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:05Z
Check the source to verify that this is the current version before use.